

One Similarity Metric Cannot Serve Every Consumer: Measured Gains, Two Preregistered Failures, and the Resolution Limit of Forecasting Them

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Abstract

Vector search systems answer “what is close” with one fixed embedding metric. The downstream model that uses the results, which we call the consumer, weighs the embedding dimensions by its own parameters, and the two notions of closeness need not agree. We rerank approximate-nearest-neighbor candidates by the consumer-induced metric $d_C(x, y)^2 = (x - y)^\top P_C(x - y)$, where $P_C = W^\top W$ is built from the consumer’s own weight matrix and nothing is trained on retrieval outcomes, and we measure what happens across sixteen consumer-corpus pairs on real moral-text and newsgroup data. Three findings. First, a conditional dissociation is real and replicates: task accuracy rises by up to +0.29 while agreement with the embedding-nearest neighbor collapses to 0.006–0.05, it appears only where the consumer is decodable and misaligned with the embedding geometry, and a matched consumer shows nothing. Second, two preregistered attempts at a quantitative law for the effect’s magnitude failed their own thresholds, and we report both failures as executed: per-unit conversion of measured headroom spans two orders of magnitude within a single preregistered batch of units, and neither published reliability weights nor cross-fitted classification headroom orders it. Third, a post-hoc study across all thirteen evaluated units identifies the magnitude in kind and shows why it resisted sealing: the gain equals how much better nearest-neighbor labels agree under d_C than under L2, at roughly unit slope (Pearson 0.91). That quantity is measurable in advance in principle, since it is itself a retrieval statistic on labeled data; what failed is resolution. At natural corpus sizes its sampling noise (0.01–0.06 across folds) is the size of most real effects, so both preregistrations had barred distinctions finer than the available data could resolve. The value of consumer-aware retrieval is real and lawful in form, and forecasting it is a statistical power problem: cheap where the effect is large, and beyond the reach of ordinary labeled corpora exactly where most natural effects live.

1 Introduction

Approximate-nearest-neighbor (ANN) retrieval fixes a metric when the index is built. FAISS ranks by L2 or inner product; production vector databases expose a short menu of global metrics and default to cosine. The application that consumes the retrieved items, the consumer in what follows, usually weighs the embedding differently: a moderation model attends to a few directions, a topic classifier to others, and neither weighting is the index metric. We refer to the consumer’s induced weighting as its read, and to $d_C(x, y)^2 = (x - y)^\top P_C(x - y)$ with $P_C = W^\top W$ as its read metric; everything in the paper reduces to these two definitions. This paper asks what happens when the ranking step uses the consumer’s read instead. Candidates are generated by the index metric as usual; the top k are reranked by the read metric. Nothing is learned against retrieval outcomes, and the index is not modified.

Three findings, in the order the evidence arrived. The dissociation is real and conditional: reranking by the read raises the consumer’s task accuracy while destroying agreement with the embedding-nearest neighbor, it requires the consumer to be both decodable from the embedding and misaligned with its geometry, and

a consumer aligned with the geometry gains nothing. Our two attempts to preregister a quantitative law for the magnitude both failed their sealed bars, and the failures are part of the record reported here. The post-hoc account of why is the paper’s sharpest result: the magnitude equals the read metric’s own retrieval headroom at roughly unit slope, and forecasting it is limited by resolution rather than by principle: at the corpus sizes natural consumers provide, the sampling noise of the forecast is as large as most of the effects. A practitioner can obtain the gain, and can measure it in advance exactly down to the resolution of the labeled data at hand, which for most of our units was not fine enough.

The components are classical, and we claim none of them. Mahalanobis and learned metrics, local metric learning, and trained rerankers are established; position-dependent metrics for exploratory data analysis go back to the learning-metrics program of Kaski and colleagues, and the view that embedding quality is relative to a user cost is explicit in NeRV. What this paper adds is the provenance of the metric, taken from the deployed consumer rather than learned for retrieval, and the measurement: the conditional dissociation with both arms sealed, the two honest failures, and the identification of the magnitude with an unforecastable potential.

2 Method

Substrate. Sentence embeddings from two frozen encoders: all-MiniLM-L6-v2 (384-d) over 20 Newsgroups for the construction sweep, and LaBSE (768-d, L2-normalized) for all graded units. Graded corpora are single-domain moral-text sets (Social-Chemistry-101 foundations and judgment, AITA privacy, hate-speech fairness, dark-pattern autonomy, Moral-Stories care, environmental claims), pinned at seal with SHA-256 manifests and row counts in the repository.

Consumers. An experimental unit is one (label axis, corpus) pair, for example fairness labels over a hate-speech corpus; each unit supplies one consumer. The consumer is a logistic probe on the embedding, $P_C = W^\top W$. Where a validated domain encoder exists for the axis, the probe is distilled from it: the encoder’s centroid-valence score labels the calibration split and the probe regresses that read onto the embedding space. Probes are trained on calibration data only, before grading data is opened.

Pipeline and metrics. FAISS IndexFlatL2 generates $k = 50$ candidates; reranking picks the top-1 by d_C ; the task scores the query’s label against the returned neighbor’s label. Reported per unit and seed: gain (reranked minus L2 accuracy), recall@1 (agreement with the L2-nearest), an isotropic control ($P_C = I$ must reproduce L2 within 0.005; it did, in every unit and seed of both families), and the oracle ceiling over the candidate set (at least 0.997 everywhere, so candidate generation never binds and every effect is the metric’s).

Discipline. We use preregistration in the confirmatory-science sense: pass/fail thresholds, units, and random seeds are frozen (sealed) in version control before any evaluation data is opened, and the verdict is committed exactly as the frozen rules grade it. Exploratory runs commit their predictions to version control before executing. Each dataset splits into a calibration part, used for training probes and for all qualification decisions, and a grading part, opened only for the final sealed evaluation. Each preregistered batch of units is a family. Units are assigned mechanically to one of two prediction arms: the misaligned arm, where the theory predicts a gain (decodability $D \geq 0.75$ and headroom $H = D - A \geq 0.04$, with A the L2 1-NN label accuracy), and the matched arm, where it predicts none ($|H| < 0.02$). Family V1 sealed seven consumers this way with three disjoint evaluation seeds. Family V2 repaired the two instrument defects V1’s post-mortem found: qualification moved to five-fold cross-fitting and every unit became single-corpus. All records, seals, and post-mortems are in the campaign repository.

3 The dissociation is real and conditional

The exploratory sweep established the shape. On synthetic embeddings with a forced off-axis label, reranking by the read moved task accuracy from 0.261 to 0.676 while recall@1 fell from 1.000 to 0.027. On real 20 Newsgroups embeddings with the consumer aligned to the task (topic), the effect vanished (-0.019): a well-trained encoder is already the topic consumer’s metric, and there is nothing to exploit. Between those

poles the effect is monotone in measured misalignment: binarizing progressively lower-variance PCA directions of the same real embeddings moves the gain from +0.08 to +0.41. Natural attributes behave accordingly (document length +0.185; capitalization +0.019). A cross-lingual arm found that an English-trained read operator transfers to machine-translated Spanish and French databases with positive but attenuated gains (+0.054, +0.033), and refuted the stronger claim that the read survives translation better than L2 does.

The graded families confirm the qualitative law wherever it was tested. Across the thirteen graded units, eleven of the anisotropic consumer-seed cells show positive gain; the two matched-arm units sit at their predicted null (V1 privacy +0.007 mean; V2 autonomy +0.0099 against a 0.010 bar); recall@1 collapses in every unit; the isotropic control never moves. The largest sealed effects are large: +0.263 to +0.292 across three seeds on the care unit, and +0.063 to +0.101 on AITA privacy.

4 Two sealed failures, reported as executed

Family V1 barred each misaligned unit at $0.3H$ per seed and each matched unit at +0.01. Verdict: FAIL. Qualifying units converted about 9% of measured headroom against the sealed 30%; the matched bar broke by 0.003 on one seed. The post-mortem found two instrument defects. The single-split headroom estimate is noisy: one consumer was excluded at $H = +0.026$ when its cross-fit headroom was $+0.080 \pm 0.014$, and it then produced the run’s largest gains (+0.053 to +0.062). And pooled two-domain corpora manufacture headroom no query population carries: one axis was matched in one domain ($D = 0.944$, $A = 0.914$) and weak in the other ($D = 0.689$), its pooled $H = 0.048$ described neither, and its gain was correctly zero.

Family V2 repaired both defects and failed anyway, for a more instructive reason. With cross-fit qualification and single-domain units, per-unit conversion of measured headroom still spanned two orders of magnitude inside one sealed family: 190% (privacy), 74% (epistemic), 9% (fairness), ≈ 0 (environmental). The rank bar failed at Spearman 0.143. One unit exceeded its measured headroom outright, which shows the qualification quantity itself is the wrong potential: classification headroom $H = D - A$ bounds what the probe classifies, not what its metric retrieves. In both families the excluded row held the strongest confirmation: the V2 care unit, excluded for a weak probe ($D = 0.673$), converted 101% of a +0.270 headroom with three tightly clustered seeds. The qualification floor built to protect the misaligned arm from weak probes twice filtered out the largest effects of the phenomenon it guards.

5 The magnitude is the read metric’s retrieval headroom, resolved only as finely as the data allows

The post-hoc study asked what does order the magnitude. Three candidates fail: the published reliability weights of the source encoders (tested as V1’s secondary bar), classification headroom (Spearman 0.181 across the thirteen units), and single-split retrieval headroom refined by cross-fitting, which made the rank correlation worse (0.352 to 0.220). The quantity that succeeds in kind is the read metric’s own retrieval headroom: $\Phi = \text{xfit}(R_C - A)$, where R_C is the 1-NN label accuracy when neighbors are chosen by d_C instead of L2, cross-fitted on calibration data. Against Φ the graded gains lie near the unit line (Pearson 0.908, slope 1.30; Figure 1), and the identification is close to definitional once stated: with the oracle ceiling at 1 and k near-global, the graded gain is the grading split’s retrieval headroom under the frozen read.

Why then did two preregistrations fail? Not because Φ is unmeasurable; it is itself a retrieval statistic, and anyone with labeled data can compute it in advance. They failed because at natural sizes the measurement’s resolution is coarser than the effects. The fold-level standard deviation of Φ is 0.011–0.058 across units, and ten of thirteen measured gains are below 0.06: both families had sealed bars finer than the instrument could resolve. The one unit whose Φ stands eleven standard deviations from zero is forecast correctly (0.201 ± 0.018 predicted, 0.274 measured). Because calibration and grading are exchangeable splits of one corpus, the remaining prediction-versus-outcome deviations are consistent with two-sided sampling noise of both quantities, and we do not claim a systematic shift. A sealable forecast therefore needs power: engineered large- Φ units, or calibration corpora roughly an order of magnitude larger, and we record that requirement rather than attempt a third seal on the present pool.

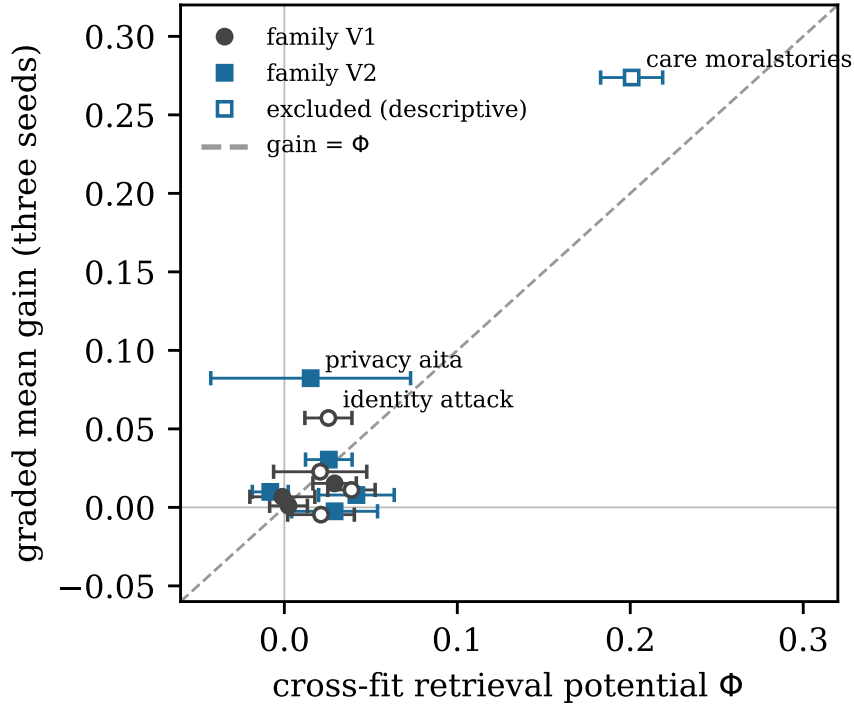


Figure 1: Graded mean gain against the cross-fit retrieval potential Φ , thirteen units, fold-SD error bars. One large- Φ unit sits on the unit line; the rest sit at the instrument’s noise floor. The correlation is Pearson 0.91 at slope 1.3, and the rank correlation is 0.22.

The finding has the same shape as the phenomenon it measures. What retrieval is worth depends on which model consumes the results; we add that how much it is worth is typically a small number, and resolving a small number in advance costs labeled data that natural consumers rarely have. The two failed preregistrations are the measurement of that resolution limit, and we regard them as results rather than mishaps.

6 Related work

Metric learning, Mahalanobis and local variants, and trained rerankers are long established; our reranker is not learned and our claim is not a better reranker. Position-dependent Riemannian metrics from auxiliary information were developed for exploratory analysis in the learning-metrics program (Kaski et al., 2001; Peltonen et al., 2004), and NeRV made embedding quality explicitly relative to a user’s precision-recall cost (Venna et al., 2010); our P_C is read off a deployed consumer rather than built from labels at embed time, and where the consumer is itself a label probe the constructions nearly coincide. Task-aware and decision-focused pipelines optimize models for downstream decisions (Elmachtoub and Grigas, 2022); sensor-selection and value-of-information literatures optimize weighted-trace objectives (Joshi and Boyd, 2009). We do not offer a better metric, reranker, or selection rule. We offer the measurement: the conditional dissociation with both arms sealed, two preregistered failures reported as executed, and the identification of the magnitude with a potential the consumer cannot estimate in advance.

7 Scope and limitations

One embedding family per phase, linear probes, top-1 label prediction, and $k = 50$ candidate generation whose ceiling never binds. The corpora are English moral text and newsgroups; the cross-lingual arm covers machine-translated Spanish and French databases only, and its stronger invariance claim is refuted above. The distillation rule ties three units to validated domain encoders; the rest use direct labels. All limits push in the same direction: they narrow the settings in which we have measured the law, not the honesty of the failures. In our controlled tests nothing suggested the dissociation appears without measured misalignment, and we claim nothing about that regime.

Reproducibility

Every number above is read from committed, hashed records in the campaign repository (sealed preregistrations, graded verdict JSONs, post-mortems, and the determinant study), and the figure regenerates from them by script. The two sealed families' verdicts are immutable; the repository's evidence discipline file describes the seal and amendment rules. An anonymized copy of the records accompanies the submission; the repository link is provided on acceptance.

References

- S. Kaski, J. Sinkkonen, and J. Peltonen, "Bankruptcy analysis with self-organizing maps in learning metrics," *IEEE Trans. Neural Networks*, vol. 12, no. 4, pp. 936–947, 2001.
- J. Peltonen, A. Klami, and S. Kaski, "Improved learning of Riemannian metrics for exploratory analysis," *Neural Networks*, vol. 17, no. 8–9, pp. 1087–1100, 2004.
- J. Venna, J. Peltonen, K. Nybo, H. Aidos, and S. Kaski, "Information retrieval perspective to nonlinear dimensionality reduction for data visualization," *JMLR*, vol. 11, pp. 451–490, 2010.
- A. N. Elmachtoub and P. Grigas, "Smart 'predict, then optimize'," *Management Science*, vol. 68, no. 1, pp. 9–26, 2022.
- S. Joshi and S. Boyd, "Sensor selection via convex optimization," *IEEE Trans. Signal Processing*, vol. 57, no. 2, pp. 451–462, 2009.